Statistical Analysis Report

Metro Interstate Traffic Volume — Capstone Project 2 (Statistical Analysis in Python)

Overview

This report presents a statistical analysis of the **Metro Interstate Traffic Volume** dataset, available from the UCI Machine Learning Repository, which records hourly westbound traffic volume on Interstate 94 near Minneapolis-St Paul, Minnesota, merged with hourly weather observations. The analysis asks whether weather conditions are associated with hourly traffic volume, and whether that relationship is better captured by a linear correlation with raw weather measurements or by a categorical comparison across weather conditions. Descriptive statistics, three visual models, and a one-way ANOVA hypothesis test were used to investigate this question in the accompanying notebook (`analysis.ipynb`).

Dataset Description

The dataset represents hourly traffic counts on I-94 collected by the Minnesota Department of Transportation between 2012 and 2018, joined with hourly weather readings sourced from OpenWeatherMap. After loading, it contains 48,204 rows and 9 columns: a holiday indicator (`holiday`), four numeric weather fields (`temp` in Kelvin, `rain_1h`, `snow_1h`, `clouds_all`), two categorical weather descriptors (`weather_main`, `weather_description`), a timestamp (`date_time`), and the numeric outcome variable, `traffic_volume`. The key variables examined in this analysis are `traffic_volume` (outcome), `weather_main` (the 11-category grouping variable used in the hypothesis test), and the four numeric weather fields used in the correlation analysis.

Methods

Before computing any statistics, an initial data analysis (IDA) pass checked data types, screened for missing values, and looked for structural issues such as duplicate timestamps. This step mirrors the practice recommended by Lusa et al. (2024), who argue that a documented, separate IDA stage rather than informal checks folded invisibly into the main analysis is necessary to build a reproducible foundation for any downstream statistical result. That check surfaced a subtle encoding issue: the `holiday` column uses the string `"None"`, which pandas parses as a missing value, to mean "not a holiday" rather than to signal genuinely absent data; this was resolved by deriving an explicit `is_holiday` flag rather than treating the column as having missing data.

Descriptive statistics (`df.describe()`, categorical value counts, and skewness) were chosen because the outcome variable and the grouping variable are of different types (continuous vs. categorical) and both needed to be characterized before any test could be chosen appropriately. Three visual models were selected to triangulate the analytical question from complementary angles: a histogram to check the overall shape of the outcome distribution, a boxplot to compare the outcome across the categorical grouping variable, and a correlation heatmap to check whether the numeric weather fields have a linear relationship with the outcome. A one-way ANOVA was then selected as the hypothesis test because the predictor of interest (`weather_main`) has more than two categories and the outcome is continuous, which rules out a two-sample t-test and makes ANOVA the standard choice for a single categorical factor with several levels (Field, 2013).

Two parametric assumptions underlie the ANOVA F-test: approximate normality of the outcome within each group, and homogeneity of variance across groups. Both were checked explicitly rather than assumed. Normality was assessed with the Shapiro-Wilk test (Shapiro & Wilk, 1965) on a fixed-size random subsample of 500 observations per weather category, since applying Shapiro-Wilk to the full sample of thousands of rows per group would flag statistically significant but practically trivial deviations from normality. Variance homogeneity was assessed with Levene's test across all included groups. Two weather categories, `Smoke` (n = 20) and `Squall` (n = 4), were excluded from the test entirely because their sample sizes are too small to estimate a within-group variance reliably.

Results

Descriptive statistics. Hourly `traffic_volume` ranges from 0 to roughly 7,280 vehicles with a mean near 3,260; `rain_1h` and `snow_1h` are heavily zero-inflated (median 0.0 in both cases), meaning precipitation hours are rare but occasionally extreme; and the `holiday` indicator is strongly imbalanced, with only 61 of 48,204 hours (about 0.1%) falling on a named holiday.

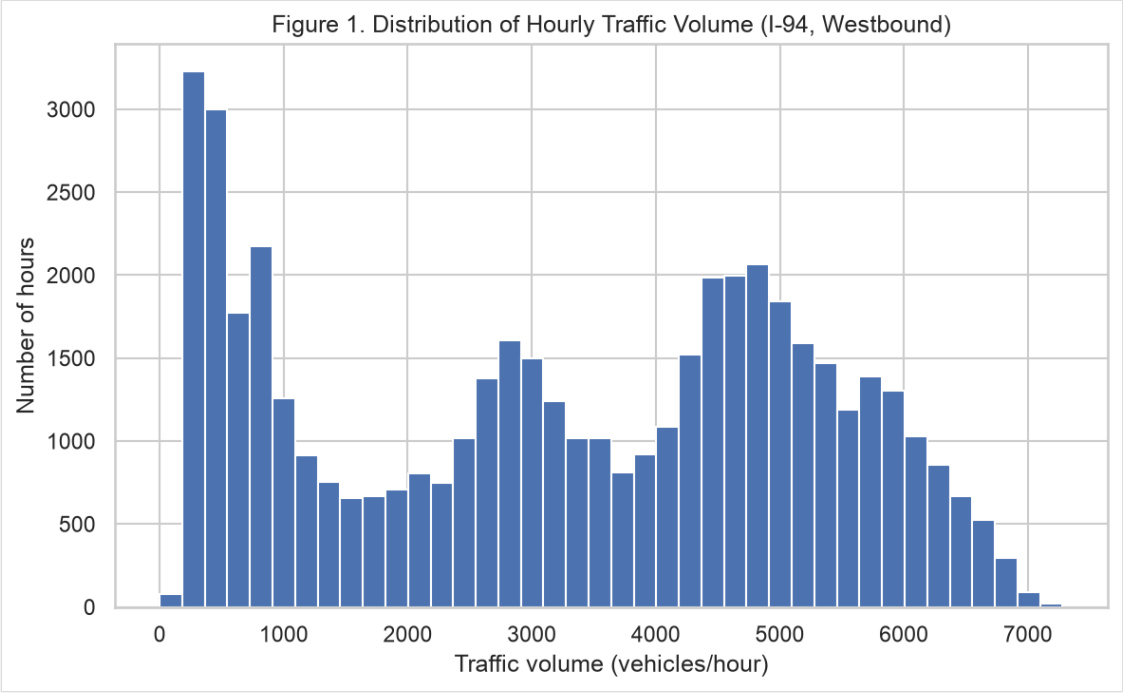


Figure 1. Distribution of hourly traffic volume. The shape is bimodal rather than normal, reflecting a mix of overnight low-traffic hours and daytime/rush-hour high-traffic hours.

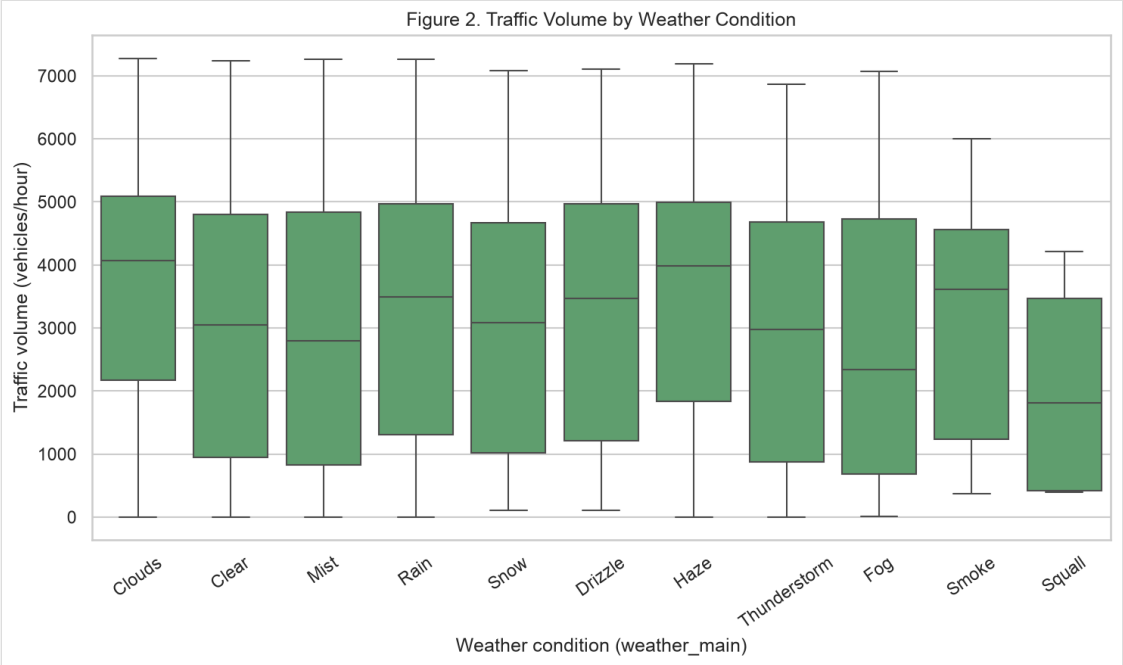


Figure 2. Traffic volume by weather condition. Medians and spread vary visibly across categories, most notably in the very small Smoke and Squall groups.

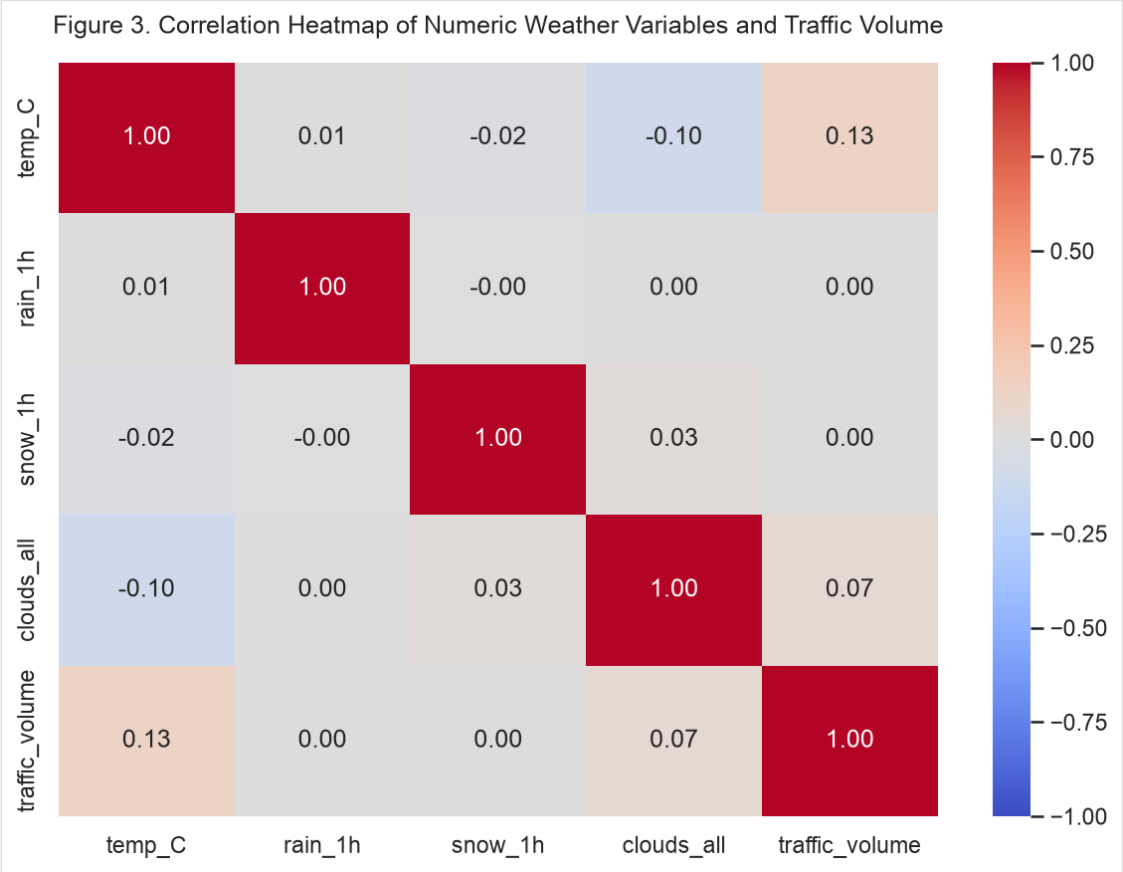


Figure 3. Correlation heatmap of the numeric weather variables and traffic volume. All correlations with traffic_volume are weak (|r| ≤ 0.13).

Hypothesis test. Null hypothesis (H0): mean `traffic_volume` is equal across all `weather_main` categories. Alternative hypothesis (H1): at least one category's mean differs from the others. After excluding `Smoke` and `Squall` (48,180 of 48,204 rows retained, 9 categories), Shapiro-Wilk tests rejected normality within every group ($p < .0001$ for all nine categories) and Levene's test rejected equal variances across groups (statistic = 39.04, $p < .0001$). The one-way ANOVA itself was statistically significant: $F(8, 48171) = 121.58$, $p < .0001$. The null hypothesis of equal mean traffic volume across weather conditions is rejected.

Because both parametric assumptions were violated, this result is interpreted as strong descriptive evidence of group separation in a very large sample rather than as a textbook-clean inferential result, a caveat consistent with the broader IDA principle that assumption checks should be reported alongside, not instead of, the headline test statistic (Lusa et al., 2024). This pattern is also consistent with Figure 3: because the numeric weather measurements show almost no linear correlation with traffic volume, the group differences detected by the ANOVA in Figure 2 appear to be a genuinely categorical effect (e.g., driven

by a small number of severe-weather hours and day-of-week/seasonal confounding) rather than one that a simple linear model on temperature, rain, snow, or cloud cover would have captured.

Interpretation for a Non-Technical Audience

In plain terms: the amount of traffic on this highway is not the same on every type of weather day, the statistical test found a real difference in average traffic levels between weather conditions like clear skies, rain, snow, and fog. However, that difference is small and uneven rather than a simple "bad weather means less traffic" rule: a handful of rare conditions (like smoke or severe squalls) had the most unusual averages, but there were too few hours of those conditions to be confident about them individually. Meanwhile, simply knowing the exact temperature, rainfall, or cloud cover for an hour tells us almost nothing on its own about how much traffic to expect — the weather "category" matters more than the weather "number." The biggest pattern in the data is not weather at all, but time of day: traffic is heavily split between quiet overnight hours and much busier daytime hours.

Limitations and Potential Bias

Dataset limitation: roughly 16% of the timestamp values in this dataset are exact duplicates, a known data-quality issue that likely originates from how the original hourly traffic feed was merged with the weather API feed. These rows were not removed in this analysis, since no documented rule existed for choosing which duplicate to keep, but their presence means some hours are implicitly over-represented in the descriptive statistics and the hypothesis test, and a more rigorous reproduction of this analysis should resolve this duplication before drawing firm conclusions (Lusa et al., 2024).

Potential source of bias: the weather categories are extremely unbalanced `Clouds` and `Clear` together account for roughly 59% of all hours, while categories such as `Smoke` (20 hours) and `Squall` (4 hours) are vanishingly rare and had to be excluded from the test outright. Any conclusion about "rare" weather conditions is therefore built on very little evidence and should not be generalized. There is also a geographic and temporal bias: the dataset reflects a single roadway in one U.S. metropolitan region over a specific six-year window, so results about weather and traffic volume here may not transfer to other cities, road types, or climates.

References

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Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika, 52*(3-4), 591–611. https://doi.org/10.1093/biomet/52.3-4.591

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